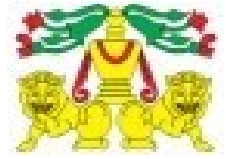




National youth Services Council
Diploma in Computer Networking & Cyber Security
Final Examination - 2026 January
Part II



Time: - 2 hours

Training Centre:-

Exam No:-.....

Answer 6 questions only and each question has 10 marks.

Question 1:

Scenario: A mid-sized company is experiencing frequent network loops and slow inter-departmental communication. You are tasked with redesigning their Layer 2 infrastructure.

- i. Explain how **Virtual LANs (VLANs)** improve network performance and security by segmenting broadcast domains within a corporate environment. **(4 Marks)**
- ii. Compare and contrast **Spanning Tree Protocol (STP)** and **Etherchannel**. Discuss how they work together to provide both redundancy and increased bandwidth without causing network loops. **(4 Marks)**
- iii. Outline a systematic **troubleshooting workflow** to identify whether a connectivity issue is occurring at the Physical, Data Link, or Network layer. **(2 Marks)**

Question 2:

Scenario: An organization is migrating from a traditional hardware-heavy routing setup to a more flexible, automated architecture to support branch offices.

- i. Discuss the advantages and disadvantages of **Dynamic Routing protocols** (like OSPF or EIGRP) over **Static Routing** in a large, frequently changing network. **(4 Marks)**
- ii. Explain the core principles of **Software Defined Networking (SDWAN)**. How does the separation of the control plane and data plane allow for better traffic management across multiple sites? **(3 Marks)**
- iii. Describe the role of **IP Security Services** (such as ACLs) in protecting network entry points from unauthorized access while maintaining high connectivity. **(3 Marks)**

Question 3:

You are migrating a local data center to **Microsoft Azure**. The requirement is to maintain high availability and secure inter-region communication.

- i. Explain how **Azure Virtual Networking** and **VNet Peering** allow for secure, low-latency communication between resources located in different Azure regions. **(3 Marks)**
- ii. Compare **Azure Virtual Machines (IaaS)** with **PaaS Compute Options**. In what scenarios would an administrator choose a PaaS solution to reduce management overhead? **(3 Marks)**
- iii. Detail the strategies for **Data Protection and Monitoring** in Azure. How do Azure Backup and Azure Monitor ensure that an organization meets its compliance and "uptime" requirements? **(4 Marks)**

Question 4

As a security analyst, you are performing a penetration test on a client's web application and internal network.

- i. Describe the "Reconnaissance" phase of a cyber-attack. How do **Information Gathering and Vulnerability Scanning** allow an attacker to map out a target's weaknesses? **(5 Marks)**
- ii. Analyze the importance of **Cryptography** in securing data. Explain the difference between symmetric and asymmetric encryption and how they are used to prevent **Password Cracking**. **(3 Marks)**
- iii. Explain the mechanics of a common **Web Exploitation** (such as SQL Injection or Cross-Site Scripting). What preventative measures should developers take to sanitize user input? **(2 Marks)**

Question 5

Scenario: You are tasked with designing a subnetting scheme for a large corporate headquarters that needs to accommodate 500 hosts per floor across 10 floors, while also planning for a separate IPv6 management network. (Use 192.168.0.0 Network for explanation)

- i. Explain the process of **Variable Length Subnet Masking (VLSM)** in IPv4. Calculate the most efficient subnet mask for a network requiring exactly 500 host addresses.
(5 Marks)
- ii. In the context of **Static and Dynamic Routing**, explain how a router's routing table handles IPv6 **Global Unicast Addresses** compared to IPv4 **Public Addresses**.
(2 Marks)
- iii. Evaluate the role of **Network Address Translation (NAT/PAT)** in IPv4. Why is NAT important
(3 Marks)

Question 6

You are given the network address 180.150.0.0; you are required to have 5 subnets.

- i. What is the minimum number of Host Bits can you take in to the Network Bits for this purpose? **(1 Marks)**
- ii. Write down the addresses of 5 subnets. (Write in binary where necessary). **(3 Marks)**
- iii. Write the subnet mask for the network. **(2 Marks)**
- iv. IPv6 (Internet Protocol Version 6) was introduced to overcome the limitations of IPv4 and to support the future growth of the Internet.

(a) Explain the structure and representation of an IPv6 address. In your answer, include address length, hexadecimal notation, and rules for shortening IPv6 addresses with suitable examples. **(2 Marks)**

(b) Describe the main types of IPv6 addresses. Explain **Unicast**, **Multicast**, and **Anycast** addresses, and briefly discuss common IPv6 address categories such as **Global Unicast**, **Link-Local**, and **Unique Local** addresses with practical use cases. **(2 Marks)**

Question 07

A server plays a critical role in modern computer networks by providing **centralized services** and resources to **users** and **client devices**.

- i. Define what a server is and explain its importance in a networked environment. **(3 Marks)**
- ii. Explain the concept of server roles in an operating system. Discuss the functions and importance of the following server roles:
 - a) Active Directory (AD) / Domain Controller (DC)
 - b) DHCP (Dynamic Host Configuration Protocol)
 - c) DNS (Domain Name System)
 - d) RDS (Remote Desktop Services)
 - e) FTP (File Transfer Protocol)**(7 Marks)**

Your answer should clearly describe how each role contributes to network management, security, and resource sharing in an organization.

Question 08

Microsoft Azure is a cloud computing platform that provides different **service models**, **deployment models**, and **resource management components** to support modern IT infrastructure.

- i. Explain the **Azure service models IaaS, PaaS, and SaaS**. Compare their responsibilities between the cloud provider and the customer, giving one example for each service model. **(4 Marks)**
- ii. Describe the **cloud deployment models: Private Cloud, Public Cloud, and Hybrid Cloud**. Explain how each model is used in organizations and discuss their advantages. **(3 Marks)**
- iii. Define the following **Azure management components** and explain their purpose in managing cloud resources: **(3 Marks)**
 - a. Tenant
 - b. Subscription
 - c. Virtual Machines (VMs)